

$$1. a) \Delta E = E \cdot \sqrt{\left(\frac{\Delta m}{m}\right)^2 + 2\left(\frac{\Delta v}{v}\right)^2}$$

$$b) \Delta g = g \cdot \sqrt{\left(\frac{\Delta M}{M}\right)^2 + 2\left(\frac{\Delta r}{r}\right)^2}$$

$$c) \Delta v = v \cdot \sqrt{3\left(\frac{\Delta r}{r}\right)^2}$$

$$d) \Delta k = k \cdot \sqrt{\left(\frac{\Delta m}{2m}\right)^2 + \left(\frac{\Delta v}{2v}\right)^2 + \left(\frac{\Delta h}{h}\right)^2}$$

$$e) \Delta f = f |e^x| \Delta x$$

$$2g) d_1 = \frac{M_1}{A} \cdot x$$

$$M_1 = (0,00279 \pm 0,00009) \text{ kg}$$

$$A = (0,01013 \pm 0,00052) \text{ m}^2$$

$$\Delta d_1 = \frac{1}{A^2} \sqrt{(A \cdot \Delta M_1)^2 + (M_1 \cdot \Delta A)^2}$$

$$\Delta d_1 = \frac{1}{(0,01013)^2} \sqrt{(0,01013 \cdot 0,00009)^2 + (0,00279 \cdot 0,00052)^2}$$

$$\Delta d_1 = 0,01697845$$

$$d_1 = \frac{M_1}{A} = 0,275419545$$

$$d_1 = (0,275 \pm 0,017) \text{ kg/m}^2 \Rightarrow (0,28 \pm 0,02) \text{ kg/m}^2$$

$$2. a) M_t = M_1 + M_2$$

$$M_t = 0,00279 + 0,1045 \text{ kg}$$

$$M_t = 0,10729 \text{ kg}$$

$$\Delta M_t = \sqrt{(\Delta a)^2 + (\Delta b)^2}$$

$$\Delta M_t = \sqrt{(0,00009)^2 + (0,0001)^2}$$

$$\Delta M_t = \sqrt{0,0000000081 + 0,00000001}$$

$$\Delta M_t = \sqrt{8,1 \times 10^{-8} + 1 \times 10^{-8}}$$

$$= \sqrt{9,1 \times 10^{-8}}$$

$$= 0,000134536$$

$$M_t = (0,10729 \pm 0,00013) \text{ kg}$$

$$(0,1073 \pm 0,0001) \text{ kg}$$

$$M_1 = 2,79 \pm 0,09 \text{ g}$$

$$M_1 = 0,00279 \pm 0,00009 \text{ kg}$$

$$M_2 = 0,1045 \pm 0,0001 \text{ kg}$$

$f(x) = ax$	$u(f) = a u(x)$
$f(x) = x^a$	$u(f) = a(x)^{a-1} u(x)$
$f(x) = e^x$	$u(f) = e^x u(x)$
$f(x) = \ln(x)$	$u(f) = \frac{1}{x} u(x)$
$f(x) = \sin(x)$	$u(f) = \cos(x) u(x)$

$f(x, y) = x + y$	$u(f) = \sqrt{u(x)^2 + u(y)^2}$
$f(x, y) = x \times y$	$u(f) = \sqrt{(y u(x))^2 + (x u(y))^2}$
$f(x, y) = x \div y$	$u(f) = \frac{1}{y^2} \sqrt{(y u(x))^2 + (x u(y))^2}$

Respostas c/ 1 Algoritmo Significativo

$$2a) (107,3 \pm 0,1) \times 10^{-3} \text{ kg}$$

$$b) (101 \pm 5) \times 10^{-3} \text{ m}^2$$

$$c) (2,8 \pm 0,2) \times 10^{-1} \text{ kg/m}^2$$

$$b) A = a \times b$$

$$A = 0,097 \times 0,10445$$

$$A = 0,01013165 \text{ m}^2$$

$$a = (9,7 \pm 0,5) \text{ cm} = 0,097 \pm 0,005 \text{ m}$$

$$b = (104,45 \pm 0,01) \text{ mm} = 0,10445 \pm 0,00001 \text{ m}$$

$$\Delta A = \sqrt{(b(\Delta a))^2 + (a(\Delta b))^2}$$

$$\Delta A = \sqrt{(0,10445 \times 0,005)^2 + (0,097 \times 0,00001)^2}$$

$$\Delta A = \sqrt{(0,00052225)^2 + (0,00000097)^2} \rightarrow 9,409 \times 10^{-5}$$

$$\Delta A = \sqrt{(0,000000273 + 0,0000000094)}$$

$$\approx 0,000522583 \rightarrow 0,00052234$$

$$A = (0,01013 \pm 0,00052) \text{ m}^2 \Rightarrow (0,0101 \pm 0,0005) \text{ m}^2$$

